

Operator norms in orthonormal coordinates

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Chapter 1

Orthonormal coordinates

1.1 Operator norms in orthonormal coordinates

Theorem 1.1.1 (Operator norm in orthonormal coordinates). [Lean](#) [✓ Stmt](#) Let E and F be inner-product spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, with finite orthonormal bases $(e_j)_{j \in J}$ and $(f_i)_{i \in I}$. For a continuous linear map $T : E \rightarrow F$, let $M_{ij} = \langle f_i, T e_j \rangle$ be its matrix in these bases. The Euclidean operator norm of M equals the operator norm of T : $\|M\|_{2 \rightarrow 2} = \|T\|$. The two basis index sets may have different cardinalities, and either may be empty.

Proof. [✓ Proof](#) Let $U : E \rightarrow \mathbb{K}^J$ and $V : F \rightarrow \mathbb{K}^I$ be the coordinate isometries. The linear map represented by M is VTU^{-1} . Composition on either side with a surjective linear isometry preserves the operator norm, so $\|M\|_{2 \rightarrow 2} = \|VTU^{-1}\| = \|T\|$. $\square$

Bibliography